PRANAW GAUTAM

AI/ML Engineer | Full-stack Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer and full-stack developer who builds machine-learning features and the production software around them: a recommender proven on held-out data (87.4% reach, p = 0.0018) and a multi-tenant school ERP SaaS in TypeScript, NestJS, Next.js and PostgreSQL. Strong in Python, SQL, evaluation against baselines, and tested, secure code from API to UI.

TECHNICAL SKILLS

Machine Learning: scikit-learn, LightGBM, TensorFlow, Recommender Systems, Collaborative Filtering, Feature Engineering, Time-Series Forecasting, Model Evaluation, Significance Testing
Languages: Python, TypeScript, JavaScript, SQL, Dart
Data & BI: Pandas, NumPy, DuckDB, Parquet, ETL Pipelines, Exploratory Data Analysis (EDA), Statistical Analysis, Power BI, DAX, Matplotlib, Seaborn
Backend & Web: NestJS, FastAPI, REST APIs, Next.js, React, zod, Tailwind CSS, Flutter
Databases: PostgreSQL (Row-Level Security), MySQL, Redis, Drizzle ORM, SQLAlchemy
Cloud, DevOps & Testing: Docker, Git, GitHub Actions (CI/CD), AWS (EC2, S3), Databricks, BullMQ, pytest, Vitest, Playwright

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Vibe6 Digital LLP, Indore **Aug 2026 – Present**

Recommendation system for a client's audio-storytelling app

- Benchmarked 8 approaches (popularity, content-based, collaborative filtering, hybrid) on time-based held-out data; designed a shelf-based home screen that reached 87.4% of held-out users vs 82.6% for a single list (McNemar p = 0.0018).
- Trained a logistic-regression ranker that lifted top-10 hit rate from 77.9% to 80.1% across 562 held-out users (p = 0.02), and enforced age gating that cut 18+ recommendations to unverified users from 99% to zero.
- Built a Flutter prototype to show the recommendations working on a phone: it ranks on-device and explains every recommendation, kept identical to the Python engine by parity tests.

Multi-tenant School ERP SaaS (TypeScript, NestJS, Next.js, PostgreSQL)

- Top contributor to main (251 of 582 commits in four weeks) on a platform with 138 tenant tables under forced row-level security: fees and online payments, transport tracking, attendance, exam marking and role dashboards.
- Built an integrations hub where each school connects its own payment gateway, GPS provider, REST APIs and HMAC-signed webhooks, with AES-256-GCM encrypted credentials, an SSRF guard and mutation-tested security checks.
- Shipped online fee payments through four gateways (Razorpay, Cashfree, PayU, Stripe) and the assessment workflow: per-student exam papers, re-evaluation requests, a consent-first webcam identity check and a 20,000-row staged import.

Data Science Intern | Shanti Infosoft LLP **Jan 2026 – Jun 2026**

- Built reusable data preprocessing pipelines in Python (Pandas, NumPy) and ran exploratory data analysis, presenting findings with Matplotlib and Seaborn.
- Debugged and optimized pipeline scripts for reliability; took part in code reviews, updates and testing across full development cycles.

PROJECTS

Vayu: Air Quality Forecasting | Python, DuckDB, LightGBM, FastAPI, React **github.com/Pranaw108/vayu-aqi**

- Normalized 71.8 million CPCB station readings from 546 files and 113 header layouts into Parquet queried through DuckDB; implemented the official CPCB AQI with 222 tests in CI.
- Built a 6-hour PM2.5 forecast with 35% lower error than the persistence baseline (MAE 41.4 vs 63.7 µg/m³) across 3 rolling-origin folds, detecting 66.5% of pollution-episode onsets.

KisanShield-FL: Crop Disease Classification | TensorFlow, FastAPI, React, Flutter **github.com/Pranaw108/kisanshield-fl**

- Reached 0.78 test macro-F1 over 12 classes and 13,618 images (EfficientNetB0); a per-source audit exposed a dataset shortcut (88.9% vs 35.0% for the same class), with grouped splits to prevent leakage.

Space Launch Analytics | Python, MySQL, SQLAlchemy, Power BI, DAX **github.com/Pranaw108/Space_Launch_Analysis**

- Built an ETL pipeline for 7,500 launches since 1957 into a 4-table MySQL warehouse, and a 6-page Power BI report on reliability, cost per kg and payload efficiency.

EDUCATION

Master of Computer Applications (MCA) | AKS University, Satna **GPA: 8.65 / 10**

Bachelor of Computer Applications (BCA) | APS University, Rewa

CERTIFICATIONS & ACHIEVEMENTS

- GOER 2025 National Finalist (top 10 in India), Institute of Risk Management (IRM) India, Jul 2025 | RMAT, IRM: 96%, May 2025
- Advanced Program on Amazon Web Services (332 hours, Grade A1), Anudip Foundation with Accenture, Mar 2025
- Data Science Bootcamp, GeeksforGeeks | Data Science Internship Certificate, Shanti Infosoft LLP, 2026